Business-RAG: Advancing Enterprise Information Extraction with Llama and DeepSeek

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Abstract. Enterprises rely on a variety of data sources, including invoices, news articles, legal documents, and financial records, to drive operations. Efficient Information Extraction (IE) is crucial for transforming this data into actionable insights to inform decision-making. Natural Language Processing (NLP) has revolutionized IE, enabling fast and precise analysis of large datasets. Techniques like Named Entity Recognition (NER), Relation Extraction (RE), Event Extraction (EE), Term Extraction (TE), and Topic Modeling (TM) are essential across industries. However, applying these techniques individually can be costly and resource-demanding, particularly for smaller organizations without significant Research and Development (R&D) resources. Large Language Models (LLMs), powered by Generative Artificial Intelligence (GenAI), provide a more affordable solution, handling multiple IE tasks simultaneously. Despite their strengths, LLMs can struggle with specialized, domain-specific queries, often leading to errors. To mitigate this limitation, Retrieval-Augmented Generation (RAG) is used to enhance LLMs by integrating external data retrieval, improving both accuracy and relevance. Although the integration of RAG with LLMs is gaining attention, its application in business settings remains relatively underexplored. This paper introduces Business-RAG, a new approach combining RAG with LLMs to improve business-related IE. We focus on Llama and DeepSeek, two open-source models that highlight the strengths of LLMs in business tasks. By integrating RAG with these models, Business-RAG offers a powerful solution for extracting actionable business insights, showcasing the potential for further exploration and improvement in business intelligence applications.

Keywords: Information extraction · Large language models · Retrieval-augmented generation · Natural language processing

1 Introduction

Enterprises rely significantly on diverse data sources, including invoices, surveys, legal documents, and more, to support their operations and make informed decisions [1]. The process of extracting valuable insights from this vast amount of data, known as

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IE, is essential for driving strategic decision-making. NLP, a branch of AI, plays a pivotal role in automating key IE tasks such as NER, RE, EE, TE, and TM [1]. NER helps identify and classify important entities, such as individuals, organizations, and locations; RE uncovers relationships between these entities; EE focuses on extracting details about events; TE identifies important terms within a given context; and TM uncovers prevalent themes or topics within a corpus of text [1]. These techniques are crucial for optimizing processes in various industries, including finance, healthcare, and manufacturing, by enabling more efficient data analysis and improving decision-making capabilities [25]. While significant research has been conducted on applying NER, RE, EE, TE, and TM in business settings, the practical implementation of these methods often requires considerable expertise and resources, which can be a barrier for many organizations [1, 25]. Rather than applying each of these methods individually, LLMs, which are grounded in NLP and powered by GenAI, offer a more comprehensive and efficient solution. GenAI, a subfield of AI, can generate a wide range of content, from text to images, through its ability to understand and synthesize vast amounts of data [21].

LLMs are trained on extensive datasets, enabling them to detect patterns and relationships between words and phrases, which enhances their ability to understand context and generate relevant, accurate information [21]. By leveraging LLMs, businesses can quickly develop AI-powered applications that streamline the extraction of valuable insights from varied sources, reducing the need for manual labor and costly model development [8, 18]. Despite their versatility, LLMs often struggle with domain-specific queries, which can lead to the generation of inaccurate or irrelevant information [10]. To overcome this limitation, RAG incorporates external data retrieval into the generative process, which boosts the accuracy and relevance of the output by providing additional context from external knowledge sources [5]. This integration allows businesses to enhance the performance of LLMs in complex, domain-specific scenarios, ensuring more reliable and contextually appropriate results. While the integration of RAG with LLMs has gained considerable attention in recent years, there remains a noticeable gap in research focusing on the development of business-related IE applications using this combination. To address this gap, this paper begins by reviewing the current body of work on Business IE and the various NLP tasks associated with it, followed by an exploration of the existing applications of RAG in conjunction with LLMs. The paper then introduces a novel application, Business-RAG, which shows the potential of integrating RAG with two distinct open-source models, Llama and DeepSeek LLMs, to develop an effective Business IE solution.

The structure of this paper is as follows: Sect. 2 provides an in-depth review of the relevant literature on Business IE, highlighting key challenges and current solutions, and explores the applications of RAG with LLMs in various domains. In Sect. 3, we present our innovative system, Business-RAG, detailing how the integration of RAG with LLMs enhances the efficiency and accuracy of IE in the business context. Section 4 offers a discussion on the advantages of the Business-RAG system, examining its performance and areas for further improvement. Finally, Sect. 5 concludes the paper, summarizing the findings and suggesting avenues for future research in this rapidly evolving area.

2 Background

IE involves the process of extracting structured data from a wide range of sources such as text documents, emails, websites, and databases [25]. This structured data consists of essential elements like entities, relationships, events, and other critical facts, which are extracted using techniques from NLP, machine learning, and computational linguistics [1, 25]. In the context of business, Business IE is specifically designed to extract actionable insights from a variety of business-relevant sources, such as financial reports, customer feedback, market research, and legal documents [28]. The extracted information can include key performance indicators (KPIs), market trends, customer sentiment, competitive analysis, and compliance data, all vital for strategic decision-making and improving business performance.

Business IE relies on five key tasks: NER, RE, EE, TE, and TM. These tasks are all rooted in NLP and aim to identify important entities, discern relationships between them, extract relevant events, identify key terms, and uncover prevalent topics within the data [25]. To accomplish these tasks, a variety of techniques are employed, including models based on Bidirectional Encoder Representations from Transformers (BERT) [4], ontology-based methods [33], machine learning algorithms [32], syntactic and semantic rules, Term Frequency-Inverse Document Frequency (TF-IDF) approaches [14], rule-based methods [12], and deep learning models [27]. These approaches help businesses automate data processing, streamline decision-making, and extract valuable insights [8].

Despite the potential benefits, there are several challenges associated with Business IE. For instance, businesses often need to rely on separate systems for each specific IE task, which can be inefficient. Additionally, choosing the right technique and ensuring the accuracy of extracted information can be difficult. Limited access to proprietary data for training models is another barrier that businesses face. These challenges highlight the need for more integrated, efficient solutions to support Business IE in practice.

Given the challenges outlined above, the potential of integrating RAG with LLMs, which are based on advanced NLP techniques, is currently being explored. LLMs are complex machine learning models that are extensively trained on large datasets to generate human-like text [23]. RAG enhances the functionality of LLMs by adding a preliminary step in which relevant external information is retrieved before text generation takes place [5]. This integration not only boosts the accuracy and relevance of the generated content but also minimizes errors and improves the overall quality of the extracted information [5].

The integration of RAG with LLMs has gained traction across various fields, improving information retrieval, content generation, and accuracy. One example is BeefBot, a chatbot for beef producers that uses LLMs to retrieve the latest agricultural insights and provide on-farm advice [9]. In the automotive industry, RAG-powered LLMs are enhancing automated design and software development by retrieving real-time technical knowledge to streamline processes [30]. Similarly, a cloud-based tool in Software Engineering education uses RAG-based LLMs to analyze student activities and provide real-time feedback, boosting UML learning outcomes [3]. Another application is an AI-powered chatbot for international graduate students, trained on curated Reddit data, which offers more accurate guidance on academic, cultural, and personal matters compared to standard models like GPT-3.5 [29]. In the legal field, a Case-Based Reasoning (CBR)-RAG approach enhances legal question answering systems by providing contextually relevant answers based on prior cases [6]. In religious Q&A systems, RAG improves response accuracy and transparency, particularly for interpreting complex texts like in the MufassirQAS system [13]. For corporate environments, RAG helps companies like Adesso Turkiye improve internal information retrieval by combining LLMs and expert-curated content, with plans for MS Teams integration [7]. In education, RAG supports personalized teaching strategies based on Pedagogical Design Patterns (PDPs), showing high accuracy [2].

In the financial sector, RAG has improved financial risk analysis by processing complex data and generating more accurate reports, emphasizing the importance of retrieval quality and prompt design [15]. RAG has also enhanced question-answering systems in programming environments by reducing hallucinations and improving accuracy and contextual sensitivity, outperforming traditional models [19]. In aviation and aerospace, RAG-powered chatbots have improved access to critical industry data, streamlining decision-making and operational efficiency [22]. Finally, in financial auditing, RAG integration has significantly enhanced the accuracy of audit report analysis, outperforming manual reviews and conventional machine learning methods, with LLaMa3.1 showing the best performance [31].

These studies highlight the wide-ranging applications and substantial advantages of integrating RAG with LLMs across various sectors, enhancing accuracy, contextual comprehension, and efficiency in fields such as agriculture, aerospace, software development, and more. However, there remains a notable gap in research focusing on the use of RAG with LLMs specifically tailored for Business IE, a gap that could greatly improve the extraction of actionable insights for business decision-making. To fill this void, the next section introduces Business-RAG [31], demonstrating how this integrated approach can be effectively applied within the business domain.

3 Business-RAG: An Integrated Solution for Business IE

3.1 Design and Implementation

To design an RAG system tailored for Business IE [31], we have leveraged core principles of EE [1, 25]. In this context, each business event represents a distinct set of named entities, intricately connected through RE techniques. Additionally, every business event encompasses a range of business-related terms, categorized under specific business topics. For example, in a Merger and Acquisition (M&A) event, the named entities might include the involved companies, executives, financial metrics, and regulatory bodies, while the RE algorithms define the relationships and implications of the transaction. Simultaneously, terms like "stock purchase agreement", "synergy", and "integration strategy" are associated with relevant domains like "corporate finance" or "strategic management". This structured framework underpins our Business-RAG system, allowing for the extraction of detailed business insights from extensive datasets.

To create an interactive platform for business insights, we developed a specialized chatbot assistant with a simple, intuitive user interface (UI) using Chainlit.io, integrated with a LLM (see Fig. 1 and Fig. 2). Our study focuses on leveraging publicly available

LLMs, specifically Deepseek-R1:1.5b [20] and Llama3.2:3b [16, 26], chosen for their compact size and exceptional performance, which can be attributed to their extensive training on diverse datasets. These models primarily serve as proof-of-concept for this application. To optimize these LLMs for Business IE tasks and derive meaningful insights from business news articles, we integrated RAG technology into our solution. This strategic combination enhances the chatbot's ability to address business-related inquiries by leveraging extracted data (See Fig. 1). The effectiveness of the chatbot is directly linked to the quality and depth of the provided business dataset. For this purpose, we used a dataset consisting of 2,845 news articles gathered from various online platforms in March 2023. Below, we examine various business-related topics that can be analyzed using the Business-RAG system (see Fig. 2), showcasing how it generates valuable business insights from news articles.

1. *Business Prospects*: By utilizing the Business-RAG framework, businesses can analyze news articles to uncover emerging market trends, identify potential investment opportunities, and track new business ventures. This includes extracting critical information about new product launches, strategic partnerships, and market expansions, offering valuable insights for decision-makers looking to capitalize on promising business prospects and making informed strategic moves.
2. *Retail Customization*: For retail businesses, the Business-RAG system enables the analysis of customer sentiment, preferences, and purchasing behaviors by examining relevant news articles. Through the extraction of data regarding consumer trends, product reviews, and shopping habits, retailers can fine-tune their product assortments, adjust pricing strategies, and develop marketing campaigns that resonate more effectively with their target customers.
3. *Financial Prediction*: Financial institutions can leverage the Business-RAG system to process and extract relevant data from news articles, providing insights that inform market trend predictions, stock performance forecasts, and broader economic indicators. By examining information from financial reports, industry analyses, and regulatory updates, businesses can enhance their ability to predict market movements and make more accurate decisions related to investments, asset management, and risk mitigation.
4. *Competitive Assessment*: The Business-RAG system also offers a powerful tool for competitive intelligence. By analyzing news articles, companies can gather valuable information about their competitors' strategies, market positioning, and industry trends. Extracted data about competitor product launches, partnerships, and acquisitions can help businesses identify areas where they can differentiate themselves, uncover new market opportunities, and recognize potential competitive threats.
5. *Customer Opinion Analysis*: The Business-RAG system empowers organizations to extract and analyze customer opinions, feedback, and reviews from business news articles, helping businesses understand customer sentiments, preferences, and concerns. By examining data on customer experiences, satisfaction levels, and perceptions of brands, companies can identify key areas for improvement, fine-tune their customer service strategies, and develop more targeted and effective marketing campaigns.

This analysis also facilitates a deeper connection with customers by allowing businesses to proactively address issues and adapt their offerings to better meet customer expectations.

6. *Regulatory Compliance and Risk Mitigation*: With Business-RAG, businesses can efficiently extract vital regulatory updates, compliance requirements, and legal developments from a wealth of news articles. This capability helps organizations stay up-to-date on changing regulations and industry standards, ensuring they meet compliance requirements and avoid legal pitfalls. By continuously monitoring legal news, businesses can stay ahead of potential risks, prevent violations, and take timely action to safeguard their operations, ultimately strengthening their risk management strategies and reducing the potential for costly legal complications.
7. *Brand Reputation Tracking*: Business-RAG offers companies a robust method to track their brand reputation by analyzing mentions of their brand, products, or services within news articles. By extracting relevant data such as sentiment analysis, media coverage, and public reactions, businesses can gain valuable insights into how their brand is perceived by the public and the media. This allows organizations to identify emerging reputation risks, address any negative publicity, and respond swiftly to any issues. Timely intervention can help businesses protect their brand image, build consumer trust, and maintain a positive public profile.
8. *Supply Chain Optimization*: Business-RAG allows organizations to extract critical supply chain data from news articles, enabling companies to enhance their logistics, reduce operational risks, and improve overall supply chain efficiency. By analyzing trends related to supplier relationships, global market shifts, and supply chain disruptions, businesses can identify potential bottlenecks and opportunities for cost savings. This data-driven approach helps companies make informed decisions related to inventory management, procurement, and distribution, ultimately optimizing supply chain strategies and ensuring smoother, more resilient operations.

Table 1 provides a summary of sample questions relevant to each business topic discussed earlier. These questions illustrate the types of inquiries that can be explored using the Business-RAG system, demonstrating its capability to extract and analyze business-related information effectively.

Table 1. Sample User Questions.

No	Business Topic	Sample Questions
1	Business Prospects	1. Are there any upcoming product launches that could impact the market? 2. What new partnerships or collaborations are emerging in the industry? 3. Which industries are experiencing significant growth or decline? 4. What potential investment opportunities are gaining attention? 5. How are market conditions expected to change in the next quarter?

(*continued*)

Table 1. (*continued*)

No	Business Topic	Sample Questions
2	Retail Customization	1. What are the latest consumer preferences and purchasing behaviors? 2. How are retailers adapting to changes in consumer sentiment? 3. Which products are receiving the most positive reviews from customers? 4. What are the current trends in customer loyalty and engagement?
3	Financial Prediction	1. What are the current predictions for stock market performance? 2. How are global economic factors affecting investment opportunities? 3. Which industries are expected to see the most significant financial growth? 4. How are changes in interest rates affecting financial markets? 5. What trends are influencing the risk factors in investment portfolios?
4	Competitive Assessment	1. Who are the top competitors in the industry and what are their strategies? 2. What new products or services are being introduced by competitors? 3. How are competitors positioning themselves in the market? 4. What recent acquisitions or mergers have occurred within the industry? 5. How are competitors adapting to new technology or market trends? 6. What market opportunities do competitors seem to be pursuing?
5	Customer Opinion Analysis	1. What are the general sentiments of customers towards our brand? 2. How do customers perceive our products compared to competitors? 3. What are the main pain points reported by customers? 4. What do customers like most about our services or products? 5. How are customer opinions evolving over time regarding our company?

(*continued*)

Table 1. (*continued*)

No	Business Topic	Sample Questions
6	Regulatory Compliance and Risk Mitigation	1. What new regulations are being introduced in our industry? 2. How are upcoming regulatory changes expected to impact on our business? 3. What legal risks do we need to prepare for soon? 4. Are there any compliance issues identified by authorities? 5. How do regulatory updates affect our risk management strategy?
7	Brand Reputation Tracking	1. How is our brand being mentioned in the media? 2. What is the general sentiment around our brand in recent news articles? 3. How do customers perceive our brand on social media platforms? 4. Are there any emerging issues that could harm our brand's reputation? 5. How are competitors' brands being discussed in the media?
8	Supply Chain Optimization	1. What disruptions are currently affecting the supply chain in our industry? 2. What are the most significant challenges in supplier relationships today? 3. How can companies mitigate risks related to supply chain disruptions? 4. What strategies are emerging to optimize logistics and delivery efficiency? 5. How are technological innovations helping businesses streamline their supply chains?

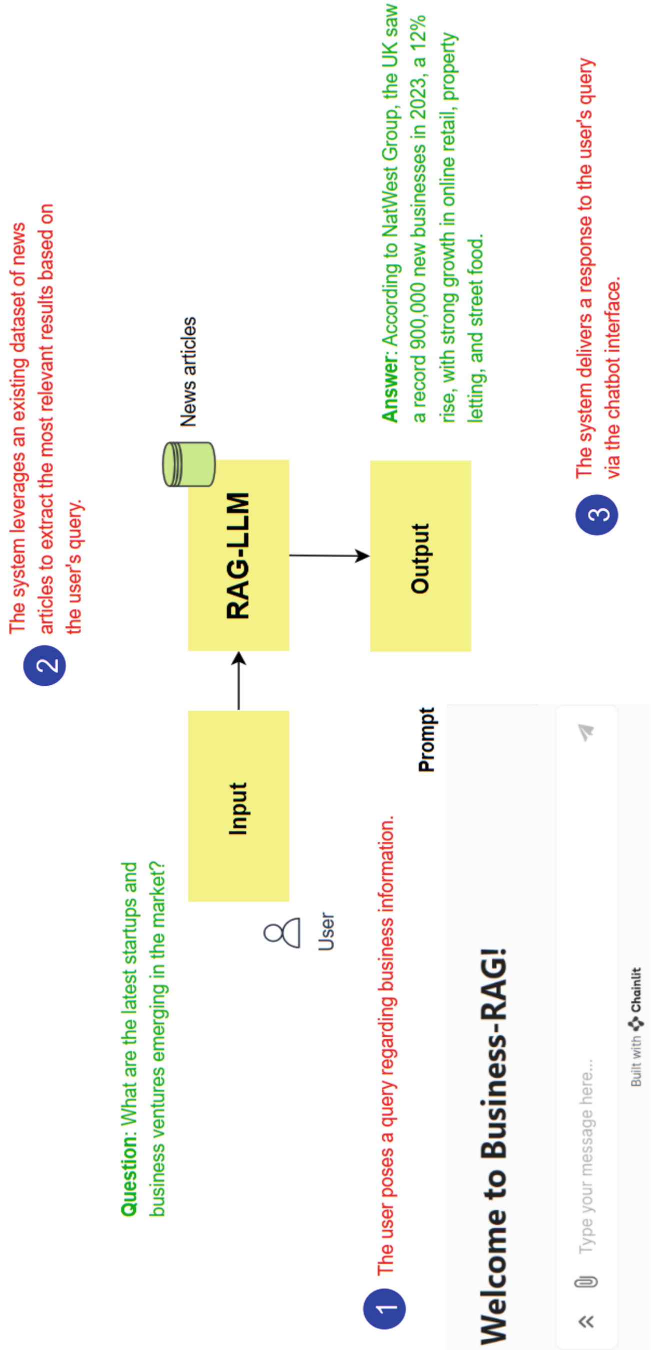

Fig. 1. Displays a visual representation of the Business-RAG application, where a business user engages with the RAG system integrated with an LLM, leveraging external business news datasets to generate relevant responses [28].

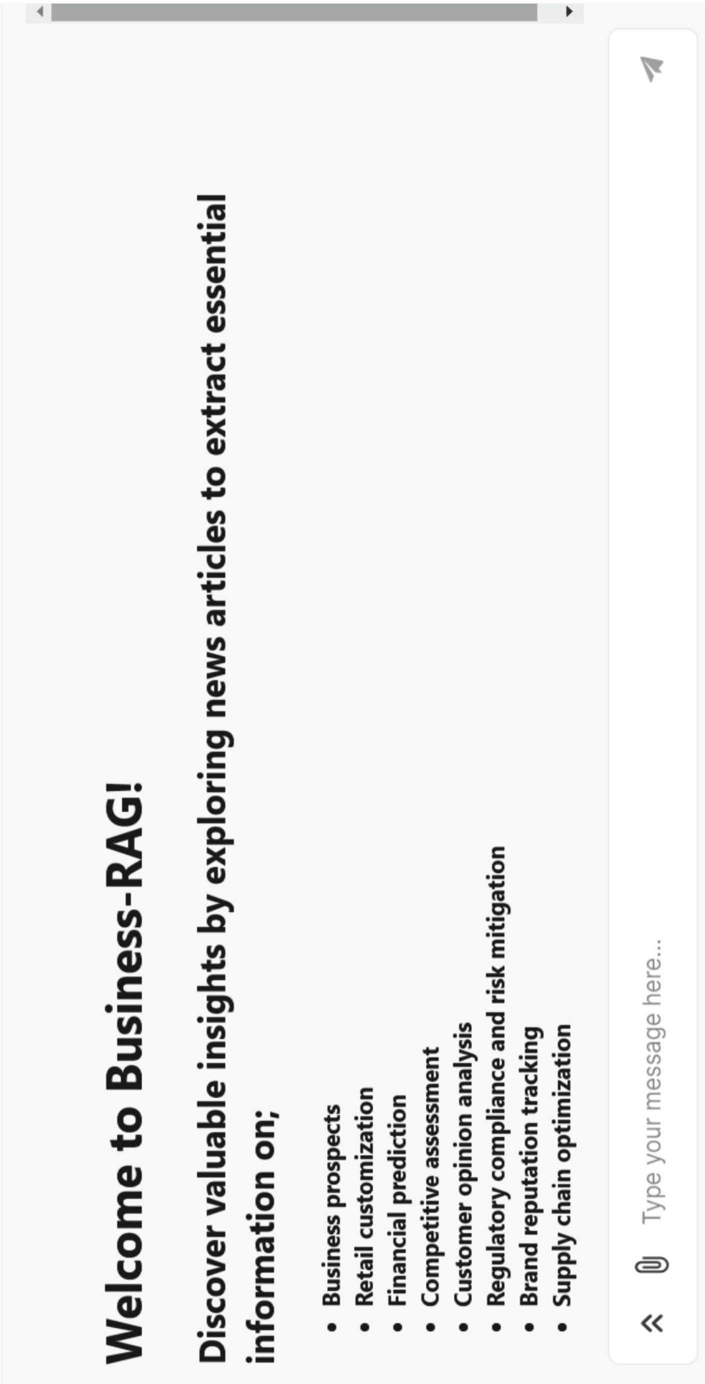

Fig. 2. User Interface of a business chatbot assistant, enabling users to seek information and insights from news article [28].

3.2 System Evaluation

To assess the system's performance, we developed a test dataset comprising 25 questions spanning various business-related topics extracted from news articles (with a selection presented in Table 2). We then analyzed the responses generated by Deepseek-R1:1.5b [20] and Llama3.2:3b [16, 26] based on several key factors to assess their effectiveness. Our evaluation focuses on accuracy of responses, examining how well each model aligns with the gold standard answers. We also consider inferences and contextual understanding, assessing each model's ability to derive implicit information from the provided context. Additionally, we evaluate their performance in handling specific information, determining how effectively they retrieve precise details. Execution time (ET) comparison is another critical factor, highlighting the efficiency of each model in generating responses. Finally, we analyze error handling, identifying how each model deals with missing or ambiguous information in the dataset. Below, we will explore these factors in detail, one by one, based on the responses recorded in Table 2.

a) **Accuracy of Responses:** Deepseek-R1:1.5b correctly answered several questions (see Table 2), including highly specific factual questions that involved numbers or precise dates, such as the building permit date for Fuji Electric and Gaëlle Morilleau's appointment as the marketing director at Cuisines AvivA. Deepseek also correctly identified the size of the warehouse for Fuji Electric and the number of new positions that SAEM Energies plans to create. In cases where it did not have the exact answer, it still tried to infer reasonable approximations, such as estimating "around thirty" for new positions, despite not being provided with precise numbers.

   Llama3.2:3b, however, tended to provide more vague or incomplete responses (see Table 2). For instance, in the case of the size of the Fuji Electric warehouse, Llama output a vastly incorrect answer of "45,757 square meters", which is not only inaccurate but also doesn't align with the context of the question. For other questions like the number of new positions SAEM Energies plans to create, Llama defaulted to "I don't know" due to the lack of explicit data in the context, which could potentially hinder its utility in real-world applications where approximate or inferred information is often valuable. The merger of Agence Wam and Première Place was another example where Deepseek-R1:1.5b provided a thorough answer, explaining that the merger's goal was to enhance their digital capabilities and increase reach. Llama3.2:3b, on the other hand, gave a less precise answer, failing to fully capture the business intentions behind the merger and instead giving a more superficial response.

   Overall, we found that Deepseek achieved an accuracy of 76.67%, while Llama demonstrated an accuracy of 65.25%.

b) **Inferences and Contextual Understanding:** One of the strengths of Deepseek-R1:1.5b lies in its ability to make reasonable inferences from the available data. For instance, in questions (see Table 2) where exact information was lacking (such as Easy Stockage's rental boxes location or MS3's building project), Deepseek still tried to build an answer based on plausible extensions of the given context. For example, when asked about Easy Stockage's location, while it couldn't find explicit details in the text, it inferred the possibility of missing information, reflecting a better contextual understanding of the gaps in the source material. On the contrary, Llama3.2:3b was more inclined to simply provide an "I don't know" answer when there was any ambiguity or missing context. While this approach may seem more conservative, it limits the usefulness of Llama in situations where even partial information could be helpful. For example, when asked about MS3's building project or the rental start date for Easy Stockage, Llama3.2:3b couldn't provide anything beyond "I don't know", which missed the opportunity to infer or approximate.

Table 2. Performance Evaluation of Deepseek-R1:1.5b and Llama3.2:3b for Business IE with Execution Time (ET).

No	Question	Gold Standard Answer	Deepseek-R1:1.5b		Llama3.2:3b	
			Answer	ET (s)	Answer	ET (s)
1	What is the size of the warehouse Fuji Electric France plans to build?	Fuji Electric France plans to build a 1,498 m^2 warehouse	The size of the warehouse Fuji Electric France plans to build is 45,757 square meters	6.9436	I don't know	1.4675
2	When did Fuji Electric France obtain the building permit?	Fuji Electric France obtained the building permit on January 11, 2023	The building permit for Fuji Electric France was obtained on January 11, 2023	4.0270	Fuji Electric France obtained the building permit on January 11, 2023	0.9467
3	What is the number of new positions SAEM Energies plans to create?	SAEM Energies plans to create 30 new positions over two years	The number of new positions created is likely several (around thirty) related to recruitment and engineering expansions	9.9957	I don't know the answer to this question based on the provided context	1.3760
4	When did SAEM acquire Cover Solar?	SAEM acquired Cover Solar in September 2022	The acquisition of Cover Solar by SAEM Energies occurred on September 2022	4.7667	According to the provided content, SAEM acquired Cover Solar in September 2022	0.9340
5	Who has been appointed as the digital marketing and communications director at Cuisines AvivA?	Gaëlle Morilleau has been appointed as the digital marketing and communications director at Cuisines AvivA	Gaëlle Morilleau has been appointed as the digital marketing and communications director at Cuisines AvivA	4.2305	Gaëlle Morilleau has been appointed as the digital marketing and communications director at Cuisines AvivA	1.1284

(*continued*)

Table 2. (*continued*)

No	Question	Gold Standard Answer	Deepseek-R1:1.5b		Llama3.2:3b	
			Answer	ET (s)	Answer	ET (s)
6	What is the main focus for Cuisines AvivA in 2023?	The main focus for Cuisines AvivA in 2023 is strengthening its digital presence	The main focus of Cuisines AvivA in 2023 appears to be responding to growth by strengthening their digital presence and expanding into new businesses	6.3966	The main focus for Cuisines AvivA seems to be strengthening its digital presence	1.3939
7	Where is Easy Stockage planning to offer 223 boxes for rental?	Easy Stockage will offer 223 boxes for rental in Distré, Loire-Atlantique, in a 1,600m^2 building and part of the former BMW garage	There is no specific information provided about where Easy Stockage offers 223 boxes for rental in the given contexts	8.4474	I don't know	0.9595
8	When will Easy Stockage begin offering the rental boxes?	Easy Stockage will begin offering 223 rental boxes in June 2023	There is no information provided in the given context that indicates when Easy Stockage will start offering the rental boxes	4.8822	I don't know when Easy Stockage will begin offering the rental boxes	0.7041
9	What is the purpose of Invers' hatchery being built in Saint-Ignat?	The hatchery being built in Saint-Ignat will enable Invers to increase its production by more than 10 times within 2 years	Invers' hatchery in Saint-Ignat aims to produce kibble for dogs and cats, improving pet welfare and providing food for local residents in Pas-de-Calais, while boosting production capacity	6.6695	I don't know	1.7672
10	When is Invers planning to start the hatchery operations?	Invers plans to start operations at the hatchery in June 2023	Invers plans to start its hatchery operations by June 2023	5.2924	Invers plans to start hatchery operations in June 2023	0.6995

(*continued*)

Table 2. (*continued*)

No	Question	Gold Standard Answer	Deepseek-R1:1.5b		Llama3.2:3b	
			Answer	ET (s)	Answer	ET (s)
11	What is the purpose of the new fundraising campaign planned by Smash?	Smash is preparing a new fundraising campaign for the second half of 2023 to support its growth and expansion, especially with the launch of a new offer for developers	The purpose of Smash's new fundraising campaign is to expand its business through expansion into new markets, enhance their product offerings, and attract more users via their subscription platform	2.5178	The purpose of the new fundraising campaign planned by Smash is to reach €15M in 2027	0.6037
12	How many customers currently use Smash's premium offer?	Smash currently has 15,000 customers subscribed to its premium offer	15,000 customers currently use Smash's premium offer	3.7140	Smash has 15,000 customers with a premium offer	0.6269
13	What is the purpose of the merger between Agence Wam and Première Place?	The merger between Agence Wam and Première Place aims to offer clients a full range of digital performance tools and expand Première Place's reach to national and international consumer brands	The purpose of the merger between Agence Wam (Lyon) and Première Place (Mulhouse) appears to be to enhance their digital capabilities, increase national and international consumer brand reach, and reinforce Agence Wam's leading position in the digital environment	6.2553	The purpose of the merger between Agence Wam and Première Place is to offer all of their clients all the levers of natural performance on the web, in order for Première Place to reach national and even international consumer brands	1.3752

(*continued*)

Table 2. (*continued*)

No	Question	Gold Standard Answer	Deepseek-R1:1.5b		Llama3.2:3b	
			Answer	ET (s)	Answer	ET (s)
14	What is the purpose of MS3's recent building project in Thuir?	MS3 has obtained a building permit to construct 661m^2 of warehouses on a 1,720m^2 plot of land in Thuir	The purpose of MS3's recent building project in Thuir appears to be the expansion of several operational and development projects across different locations within the Seine-et-Marne region	5.6718	I don't have any information about a "MS3" or a recent building project in Thuir	1.2024
15	When did MS3 receive the building permit for the warehouse project?	MS3 received the building permit for the warehouse project on December 15, 2022	Based on the provided context, there is no explicit mention of when MS3 received a building permit for their warehouse project	8.2514	I don't know when MS3 received the building permit for the warehouse project	1.5498

c) **Handling Specific Information:** A significant aspect of the evaluation is how each model dealt with specific details like dates, company names, and project specifics: Deepseek-R1:1.5b was able to correctly recognize the exact dates in several instances (see Table 2), such as the MS3 building permit date and the start of operations at Invers' hatchery. Deepseek's responses for these questions were aligned closely with the provided context and the gold standard answers. Llama3.2:3b also correctly answered several date-based queries such as the acquisition date for Cover Solar and the building permit date for Fuji Electric, showing that it could accurately answer questions with specific dates. However, the model tended to have issues with more complex queries, often defaulting to "I don't know" when it couldn't fully capture the details from the context.

d) **Execution Time Comparison:** Execution time is a key factor that can impact the usability of a model in time-sensitive applications. Llama3.2:3b excelled in speed (see Table 2), with responses often taking less than a second, making it more efficient for applications where rapid retrieval of answers is crucial. For instance, in answering questions like the appointment of Gaëlle Morilleau or the acquisition of Cover Solar, it processed the information swiftly.

   Deepseek-R1:1.5b, on the other hand, took significantly more time per response (see Table 2). For instance, it took 6–9 s on average to answer each query, which might be considered slow for real-time applications. However, this delay was generally the result of providing more accurate, well-structured, and contextually detailed responses. Deepseek took the time to process the information more deeply, leading to more complete and nuanced answers, as seen in cases like the purpose of Smash's new fundraising campaign or Invers' hatchery's role.

e) **Error Handling:** Error handling, or how the models dealt with uncertain or missing information, revealed important differences: Deepseek-R1:1.5b took a more proactive approach, even in cases where some information was lacking. It would often give an approximation or note that the exact information couldn't be found but still offers a reasonable guess or context-based interpretation. For instance (see Table 2), when answering SAEM Energies' new positions, Deepseek stated that the number was likely "around thirty" and related it to the expansion efforts. This indicated a more interpretative approach. Llama3.2:3b, in contrast, was more direct and conservative. When uncertain, it simply responded with "I don't know", such as in the case of the number of new positions at SAEM or location of Easy Stockage's rental boxes. While this approach is valid in ensuring transparency, it limits the model's ability to handle situations where approximate or inferred answers might still be useful.

4 Discussion

Extracting relevant business insights from unstructured text presents a significant challenge due to the inherent variability in data formats, structures, and contextual nuances [24]. Addressing this complexity requires the integration of multiple IE tasks, including NER, RE, EE, TE, and TM. Each of these tasks plays a crucial role in transforming raw textual data into structured business intelligence. Traditionally, organizations have relied on separate systems or tools for each of these IE tasks, leading to fragmented workflows, increased operational complexity, and resource-intensive management [1, 25]. However, recent advancements in GenAI and LLMs present a compelling alternative [17]. LLMs offer broad language understanding capabilities and the ability to process vast amounts of text, making them well-suited for Business IE.

Although LLMs are not inherently domain-specific, they can be optimized for business applications by integrating RAG. This approach enhances generative capabilities by introducing an initial information retrieval step, allowing the model to ground its responses in relevant external data sources. The Business-RAG framework leverages this combination, enabling adaptability, dynamic updates, and transparent source attribution, enhancing both the credibility and reliability of extracted business insights. To assess the effectiveness of our Business-RAG system, we conducted evaluations using two open-source models: Deepseek-R1:1.5b [20] and Llama3.2:3b [16, 26]. These models were selected to highlight performance variations across different architectures and parameter sizes. Our analysis focused on key factors such as accuracy, contextual understanding, efficiency, and execution time. The strengths for each model are outlined below:

Strengths of Deepseek-R1:1.5b: Deepseek-R1:1.5b demonstrated strong performance in accuracy, consistently delivering precise and detailed responses. Even when exact information was unavailable, it exhibited strong reasoning abilities, making logical inferences to provide likely answers. The model also excelled in contextual understanding, effectively capturing nuanced details and comprehending broader business contexts. This made it particularly reliable in scenarios where information was incomplete or ambiguous. Additionally, Deepseek maintained a high level of consistency in generating meaningful responses, reducing instances of vague or misleading outputs.

Strengths of Llama3.2:3b: Llama3.2:3b demonstrated a significant advantage in response speed, consistently outperforming Deepseek in execution time, making it a more suitable option for applications requiring rapid answers. Unlike Deepseek, which attempted to infer missing information, Llama adopted a more cautious approach by explicitly stating "I don't know" (see Table 2) when uncertain, reducing the risk of generating misleading outputs. Additionally, for straightforward factual queries, Llama proved to be highly efficient, with minimal processing overhead allowing for quick and reliable execution.

5 Conclusions

The integration of RAG with LLMs marks a significant advancement in IE, especially in the context of business applications. By leveraging LLMs, this approach provides a cost-effective and efficient solution for performing multiple IE tasks simultaneously. Despite the growing interest in combining RAG with LLMs, the literature lacks in-depth studies focusing on business-specific use cases. This paper addresses this gap by reviewing existing research on Business IE tasks and exploring the potential applications of RAG and LLMs within the business domain. Furthermore, it introduces a novel real-world application that demonstrates the practical implementation of this integration for Business IE. Using Deepseek and Llama models as proof of concept, we were able to highlight the performance differences between models from distinct families and of varying sizes. While Deepseek-R1:1.5b outperformed Llama3.2:3b in accuracy and reasoning, Llama3.2:3b excelled in speed and efficiency. The findings emphasize that the choice of model should be aligned with specific business requirements, as Deepseek is better suited for complex and in-depth analysis, while Llama is more appropriate for fast and simple queries. In future work, it will be essential to explore further improvements to the RAG-LLM integration, particularly focusing on refining model performance for business-related tasks. Future research could examine the scalability of such systems across different industries and identify ways to enhance model robustness in handling diverse and complex datasets. Additionally, integrating newer, domain-specific models and investigating the incorporation of real-time data sources would add significant value to Business IE applications. Further exploration into optimizing the balance between accuracy, speed, and resource efficiency will be vital to make these systems more adaptable to varied use cases in dynamic business environments.

References

1. Abdullah, M.H.A., Aziz, N., Abdulkadir, S.J., Alhussian, H.S.A., Talpur, N.: Systematic literature review of information extraction from textual data: recent methods, applications, trends, and challenges. IEEE Access **11**, 10535–10562 (2023)

2. Ahmed, M., et al.: CodeQA: advanced programming question-answering using LLM agent and RAG. In: 2024 6th Novel Intelligent and Leading Emerging Sciences Conference (NILES), pp. 494–499. IEEE, October 2024
3. Alan, A.Y., Karaarslan, E., Aydin, Ö.: A rag-based question answering system proposal for understanding Islam: Mufassirqas LLM. arXiv preprint arXiv:2401.15378 (2024)
4. Almeida Bordignon, A.C., Thom, L.H., Silva, T.S., Dani, V.S., Fantinato, M., Ferreira, R.C.B.: Natural language processing in business process identification and modeling: a systematic literature review. In: Proceedings of the XIV Brazilian Symposium on Information Systems, pp. 1–8, June 2018
5. Ardimento, P., Bernardi, M.L., Cimitile, M.: Teaching UML using a RAG-based LLM. In: 2024 International Joint Conference on Neural Networks (IJCNN), pp. 1–8. IEEE, June 2024
6. Arendarenko, E., Kakkonen, T.: Ontology-based information and event extraction for business intelligence. In: Ramsay, A., Agre, G. (eds.) Artificial Intelligence: Methodology, Systems, and Applications, AIMSA 2012. LNCS, vol. 7557, pp. 89–102. Springer, Berlin, Heidelberg (2012). https://doi.org/10.1007/978-3-642-33185-5_10
7. Arslan, M., Cruz, C.: Business text classification with imbalanced data and moderately large label spaces for digital transformation. Appl. Netw. Sci. **9**(1), 11 (2024). https://doi.org/10.1007/s41109-024-00623-5
8. Arslan, M., Cruz, C.: Business-RAG: information extraction for business insights. In: 21st International Conference on Smart Business Technologies, pp. 88–94. SCITEPRESS-Science and Technology Publications, July 2024. https://doi.org/10.5220/0012812800003764
9. Arslan, M., Cruz, C.: Extracting business insights through dynamic topic modeling and NER. In: KDIR, pp. 215–222 (2022). https://doi.org/10.5220/0011552900003335
10. Arslan, M., Ghanem, H., Munawar, S., Cruz, C.: A survey on RAG with LLMs. Procedia Comput. Sci. **246**, 3781–3790 (2024)
11. Arslan, M., Munawar, S., Cruz, C.: Business insights using RAG–LLMs: a review and case study. J. Decis. Syst., 1–30 (2024)
12. Bellan, P., Dragoni, M., Ghidini, C.: Extracting business process entities and relations from text using pre-trained language models and in-context learning. In: Almeida, J.P.A., Karastoyanova, D., Guizzardi, G., Montali, M., Maggi, F.M., Fonseca, C.M. (eds.) Enterprise Design, Operations, and Computing, EDOC 2022. LNCS, vol. 13585, pp.182–199. Springer, Cham (2022). https://doi.org/10.1007/978-3-031-17604-3_11
13. Bzhalava, L., Kaivo-oja, J., Hassan, S.S.: Digital business foresight: keyword-based analysis and CorEx topic modeling. Futures **155**, 103303 (2024)
14. Darji, A., Kheni, F., Chodvadia, D., Goel, P., Garg, D., Patel, B.: Enhancing financial risk analysis using RAG-based large language models. In: 2024 3rd International Conference on Automation, Computing and Renewable Systems (ICACRS), pp. 754–760. IEEE, December 2024
15. Dehbozorgi, N., Kunuku, M.T., Pouriyeh, S.: Personalized pedagogy through a LLM-based recommender system. In: Olney, A.M., Chounta, IA., Liu, Z., Santos, O.C., Bittencourt, I.I. (eds.) Artificial Intelligence in Education. Posters and Late Breaking Results, Workshops and Tutorials, Industry and Innovation Tracks, Practitioners, Doctoral Consortium and Blue Sky, AIED 2024. CCIS, vol. 2151, pp. 63–70. Springer, Cham (2024). https://doi.org/10.1007/978-3-031-64312-5_8
16. Devlin, J., Chang, M.W., Lee, K., Toutanova, K.: Bert: pre-training of deep bidirectional transformers for language understanding. arXiv preprint arXiv:1810.04805 (2018)
17. Dubey, A., et al.: The llama 3 herd of models. arXiv preprint arXiv:2407.21783 (2024)
18. Feuerriegel, S., Hartmann, J., Janiesch, C., Zschech, P.: Generative AI. Bus. Inf. Syst. Eng. **66**(1), 111–126 (2024)
19. Guo, D., et al.: Deepseek-r1: Incentivizing reasoning capability in LLMs via reinforcement learning. arXiv preprint arXiv:2501.12948 (2025)

20. Kandpal, N., Deng, H., Roberts, A., Wallace, E., Raffel, C.: Large language models struggle to learn long-tail knowledge. In: International Conference on Machine Learning, pp.15696–15707. PMLR, July 2023
21. Korger, A., Baumeister, J.: Rule-based semantic relation extraction in regulatory documents. In: LWDA, pp. 26–37, September 2021
22. Lewis, P., et al.: Retrieval-augmented generation for knowledge-intensive NLP tasks. In: Advances in Neural Information Processing Systems, vol. 33, pp. 9459–9474 (2020)
23. Martinez-Rodriguez, J.L., Hogan, A., Lopez-Arevalo, I.: Information extraction meets the semantic web: a survey. Semant. Web **11**(2), 255–335 (2020)
24. Raiaan, M.A.K., et al.: A review on large language models: architectures, applications, taxonomies, open issues and challenges. IEEE Access **12**, 26839–26874 (2024)
25. Saha, B., Saha, U.: Enhancing international graduate student experience through AI-driven support systems: A LLM and RAG-based approach. In: 2024 International Conference on Data Science and Its Applications (ICoDSA), pp. 300–304. IEEE, July 2024
26. Şahin, G., Varol, K., Pak, B.K.: LLM and RAG-based question answering assistant for enterprise knowledge management. In: 2024 9th International Conference on Computer Science and Engineering (UBMK), pp. 1–6. IEEE, October 2024
27. Sun, T.: Relation extraction from financial reports. Doctoral dissertation, University of York (2022)
28. Touvron, H., et al.: Llama 2: open foundation and fine-tuned chat models. arXiv preprint arXiv:2307.09288 (2023)
29. Wiratunga, N., et al.: CBR-RAG: case-based reasoning for retrieval augmented generation in LLMs for legal question answering. In: Recio-Garcia, J.A., Orozco-del-Castillo, M.G., Bridge, D. (eds.) Case-Based Reasoning Research and Development, ICCBR 2024. LNCS, vol. 14775, pp. 445–460. Springer, Cham (2024). https://doi.org/10.1007/978-3-031-63646-2_29
30. Yadav, S.: AeroQuery RAG and LLM for aerospace query in designs, development, standards, certifications. In: 2024 IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT), pp. 1–6. IEEE, July 2024
31. Zhang, Z., Wilson, C.A., Hay, R., Everingham, Y., Naseem, U.: BeefBot: harnessing advanced LLM and RAG techniques for providing scientific and technology solutions to beef producers. In: Proceedings of the 31st International Conference on Computational Linguistics: System Demonstrations, pp. 54–62, January 2025
32. Zhao, Y., et al.: Optimizing LLM based retrieval augmented generation pipelines in the financial domain. In: Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 6: Industry Track), pp. 279–294, June 2024
33. Zolfaghari, V., Petrovic, N., Pan, F., Lebioda, K., Knoll, A.: Adopting RAG for LLM-aided future vehicle design. In: 2024 2nd International Conference on Foundation and Large Language Models (FLLM), pp. 437–442. IEEE, November 2024